Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*		Continuous spectrum due to radiation of all wavelengths emitted from surface of star [1] [superimposed] line absorption spectrum (due to passage of radiation) through atmosphere (of star) [1]	2			2		
	(b)	(i)	Use of $I = \frac{P}{4\pi R^2}$ [1] Substitution: $\frac{I_S}{I_V} = \frac{9.7\times10^{27}\times[4\pi]\left(2.4\times10^{17}\right)^2}{1.5\times10^{28}\times[4\pi]\left(8.1\times10^{16}\right)^2}$ [1] **Alternative for 2nd mark:** Calculation of either intensity: $I_{Sirius} = 1.18 \times 10^{-7}$ [W m^{-2}] or $I_{Vega} = 2.07 \times 10^{-8}$ [W m^{-2}] Ratio = 5.7 [1]	1	1 1		3	2	
		(ii)	Shape of curve below that of Sirius at all points [1] Peak at (1,λ_{max}) **ecf** [1]		2		2		
		(iii)	Substitution into $P = 4\pi R^2 \sigma T^4$ i.e. $9.7 \times 10^{27} = 1.8 \times 10^{19} \times 5.67 \times 10^{-8} \times T^4$ [1] $T = 9.9 \times 10^3$ K (9 874 K) [1] Substitution into $\lambda_{max} = \frac{W}{T} = \frac{2.90\times10^{-3}\,[\text{m K}]}{9.9\times10^3\,[\text{K}]}$ **ecf** [1] $\lambda_{max} = 2.9 \times 10^{-7}$ [m] [1]	1 1	1 1		4	4	